Surrogate models for Monte Carlo uncertainty propagation with few runs of an expensive model: specification, selection and a valid surrogate contribution

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Data and code availability

All code, the per-seed result records, the cached CFD runs and the scripts that regenerate every table, figure and quoted number are openly available at `https://github.com/Yugam2508/fyp-active-learning-cfd`, archived at Zenodo (release v1.0), DOI 10.5281/zenodo.23661235.

CRediT authorship contribution statement

Yugam Jinesh Vora: Conceptualization, Methodology, Software, Investigation, Formal analysis, Visualization, Writing – original draft. **Wee Hoe Ng:** Conceptualization, Supervision, Writing – review and editing. [CONFIRM]

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. [CONFIRM]

Acknowledgements

[CONFIRM: e.g. NUS Engineering Science Programme, National Metrology Centre; or none.]

Prior presentation

The study at a budget of 20 runs on four models was submitted as an extended abstract to the IMEKO XXV World Congress (2027, TC21) [CONFIRM status and reference at submission]; the present manuscript extends it to twelve models, budgets of 10–80 runs, Bayesian and recalibrated GP intervals, a cross-check against an independent implementation, a failure mode, dependent inputs, vector outputs and a worked example with an uncertainty budget.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

[CONFIRM — draft; must describe the actual use truthfully.] During this work the authors used Claude (Anthropic), an AI assistant, to write and test the analysis and simulation code, run the computational experiments, check the formulas and references, and draft parts of the manuscript text. All code, results and text were reviewed and verified by the authors; every number in the manuscript is regenerated from the result records by a published script. The authors take full responsibility for the content of the published article.